

CMSE 801 - Day 3

Lists and loops

Sep 12, 2022

General Course Announcements

- Homework 1 has been released and is due Wednesday Sep 28 at 11:59 p.m.
- My slides are available on D2L

Questions/comments from Day 3 PCA

- I am confused that in some other programming language, there is some "end" words to separate the inside and outside of the loop, but here seems to have no such word. How can we define whether the line belongs to the loop or not?
- why do we start counting at 0? lists can be used like vectors?
- I think more discussion of part 3 is warranted. I attended Jim's office hours and things were much clearer after that, but a step-by-step explanation of how to have a for loop carry out more complex functions
- I think, I struggled in creating a variable from the "for" loop. I would get the output I wanted, but was unsure of storing.
- Go over more loops / my 2 variable for loop didn't work / maybe just explaining loops in classes again and how for and while loop commands are structured / More discussion of combining lists and loops. Doing so was very confusing for me and I felt lost / how do I make the 2 variable for loop work
- For a composite lists, is there a way to flatten them directly like the function ArrayFlatten in mathematica?
- I wish this had included a list comprehension example as these are very powerful.
- Let me know if I need to say more than what I am putting in this form.
- Can we move Monday pre-class assignment deadline to Saturday, I have to do research at workdays and don't have enough time for assignment. it takes me four to five hours for each assignment / The deadline of pre class assignment is a little bit tight for me.
- Is it okay to use tools not yet taught in class to solve assignments/homework?

```
In [1]: # list
Name = list()
print(Name)
Name.append('Yann')
print(Name)
Name.append('Meing')
Name.append('Law')
Name.append('Allo')
```

```
Name.append('Law')
print(Name)
Name_1 = ['Yann']
print(Name_1)
Name_1.append('3')
print(Name_1)
```

```
[]
['Yann']
['Yann', 'Meing', 'Law', 'Allo', 'Law']
['Yann']
['Yann', '3']
```

```
In [2]: print(Name)
Name.remove('Law')
print(Name)
Name.insert(2, 'Good')
print(Name)
```

```
['Yann', 'Meing', 'Law', 'Allo', 'Law']
['Yann', 'Meing', 'Allo', 'Law']
['Yann', 'Meing', 'Good', 'Allo', 'Law']
```

```
In [3]: # for loop by index
grade = [9, 10, 7, 8, 9]
info = list()
print('len=', len(Name))
for i in range(len(Name)):
    student_info = list()
    student_info.append(Name[i])
    student_info.append(grade[i])
    info.append(student_info)
print(info)
print(info[0])
print(info[0][1])
print(len(info[0]))
```

```
len= 5
[['Yann', 9], ['Meing', 10], ['Good', 7], ['Allo', 8], ['Law', 9]]
['Yann', 9]
9
2
```

```
In [4]: # for loop by value
print(grade)
average = 0
for i in grade:
    print(i)
    average += i # average = average + i
average /= len(grade) # average = average/len(grade)
print(average)
```

```
[9, 10, 7, 8, 9]
9
10
7
8
9
8.6
```

```
In [5]: for i in info:
        print(i)
```

```
['Yann', 9]
['Meing', 10]
['Good', 7]
['Allo', 8]
['Law', 9]
```

```
In [6]: for i in info:
        print(i[0], 'obtains', i[1])
```

```
Yann obtains 9
Meing obtains 10
Good obtains 7
Allo obtains 8
Law obtains 9
```

```
In [7]: # while loop
student = 'Law'
sdum = ''
i = -1
while sdum != student:
    i += 1
    sdum = info[i][0]
print(i)
print(student, 'obtains', info[i][1])
```

```
4
Law obtains 9
```

```
In [8]: x = [2,4,6,8,10,12,14,16,18]
x2 = list()
for i in x:
    x2.append(i**2)
print(x2)
```

```
[4, 16, 36, 64, 100, 144, 196, 256, 324]
```

```
In [9]: # nested loop
print(info)
for i in range(4):
    print('---')
    print('i = ', i)
    print('---')
    for j in range(2):
        print('    j = ', j)
        print('        info[i][j] =', info[i][j])
```

```
[['Yann', 9], ['Meing', 10], ['Good', 7], ['Allo', 8], ['Law', 9]]
```

```
---
```

```
i = 0
```

```
---
```

```
    j = 0
```

```
        info[i][j] = Yann
```

```
    j = 1
```

```
        info[i][j] = 9
```

```
---
```

```
i = 1
```

```
---
```

```
    j = 0
```

```
        info[i][j] = Meing
```

```
    j = 1
```

```
        info[i][j] = 10
```

```
---
```

```
i = 2
```

```
---
```

```
j = 0
    info[i][j] = Good
j = 1
    info[i][j] = 7
---
i = 3
---
j = 0
    info[i][j] = Allo
j = 1
    info[i][j] = 8
```

```
In [10]: grade_copy = grade
print(grade)
print(grade_copy)
grade[1] = 100
print(grade)
print(grade_copy)
grade_copy[2] = -1
print(grade)
```

```
[9, 10, 7, 8, 9]
[9, 10, 7, 8, 9]
[9, 100, 7, 8, 9]
[9, 100, 7, 8, 9]
[9, 100, -1, 8, 9]
```